# Generative AI-enhanced multiagent deep reinforcement learning for cooperative odor source localization

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**Abstract**

Indoor odor source localization (OSL) plays a critical role in public safety missions and environmental monitoring. Conventional single-robot localization methods suffer performance degradation in dynamic and complex environments, which has motivated a paradigm shift toward multi-robot collaborative localization. In addition, traditional robot localization methods typically exploit only a single time-varying characteristic of the source release and are therefore prone to failure in turbulent environments. To solve these problems, this paper proposes a diffusion model-enhanced multiagent proximal policy optimization (DM-MAPPO) algorithm. First, considering that an OSL task exhibits dual time-varying characteristics of periodicity and attenuation, it is modeled as a decentralized partially observable Markov decision process (Dec-POMDP) with unknown transition probabilities. Second, inspired by generative artificial intelligence (GAI), we integrate the inference and generative capabilities of diffusion models into policy learning so as to enable optimal decision-making. Finally, computational fluid dynamics (CFD) simulations are conducted in an environment with obstacles, and the results show that the proposed DM-MAPPO algorithm outperforms the four baseline algorithms.

*Keywords:* Odor source localization (OSL), Diffusion model (DM), Generative artificial intelligence (GAI), Multiagent deep reinforcement learning (MADRL), Proximal policy optimization (PPO).

## 1. Introduction

Indoor odor source localization (OSL) is an essential component of environmental monitoring and public safety missions, with representative applications including chemical gas leak detection

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in industrial facilities [1], post-disaster rescue operations under toxic gas contamination [2], and indoor environmental safety inspection [3]. In the uncertain environmental observations, achieving rapid and accurate leak source localization using autonomous systems has attracted increasing research interest. As illustrated in [4], the autonomous search process is typically divided into three sequential phases: (i) plume finding, robots situated in low-concentration regions perform a random search to find the plume; (ii) plume tracking, robots exploit the spatial distribution characteristics of the plume concentration together with historical information to navigate toward high-concentration regions; and (iii) plume localization, a robot declares the estimated leak source location once the measured concentration satisfies a prescribed threshold.

During autonomous indoor search, sensor-equipped robots seek the source by perceiving gas concentration and wind information. This process is confronted with three major challenges, as illustrated in Figure 1. The first challenge lies in the inefficiency of plume discovery through random search. Robots operating in low-concentration or even plume-free regions generally follow zigzag, spiral, or completely random search patterns [5]. Under special circumstances, such as slow plume diffusion, corner locations where plume information is persistently unavailable, and paths without plume or wind information, robots require a longer time to discover the plume. The second challenge stems from local recirculation zones and local concentration extrema, which restrict the source-searching capability of the robots. Indoor environments generally contain more corners and denser obstacles than outdoor spaces, causing wind and concentration information to exhibit local extrema. As a result, robot sensors may even fail to perceive valid airflow and concentration information. How a robot can avoid or escape such regions therefore remains a significant challenge. The third challenge arises in sparse and intermittent plume environments, where robots can hardly acquire valid information during plume tracking. When robot sensors experience missing samples and false peaks, the incomplete information makes stable convergence of concentration gradient tracking difficult. Meanwhile, the limited availability of reliable observations forces robots to rely on historical trajectories and sporadically triggered events, which increases decision-making uncertainty and prolongs the overall search time.

Owing to its strong adaptability and decision-making capability in dynamic environments [6], deep reinforcement learning (DRL) has emerged as a reliable solution to the autonomous odor source localization (OSL) problem. Guided by reward signals, the agent learns decision-making policies in different environments and updates its current policy using past experiences, with the ultimate goal of maximizing the cumulative reward [7]. In the OSL problem, the presence of obstacles and walls increases environmental complexity, while the sparse and intermittent nature of the plume renders the data collected by robots unstable [8]. As a result, OSL constitutes a sequen-

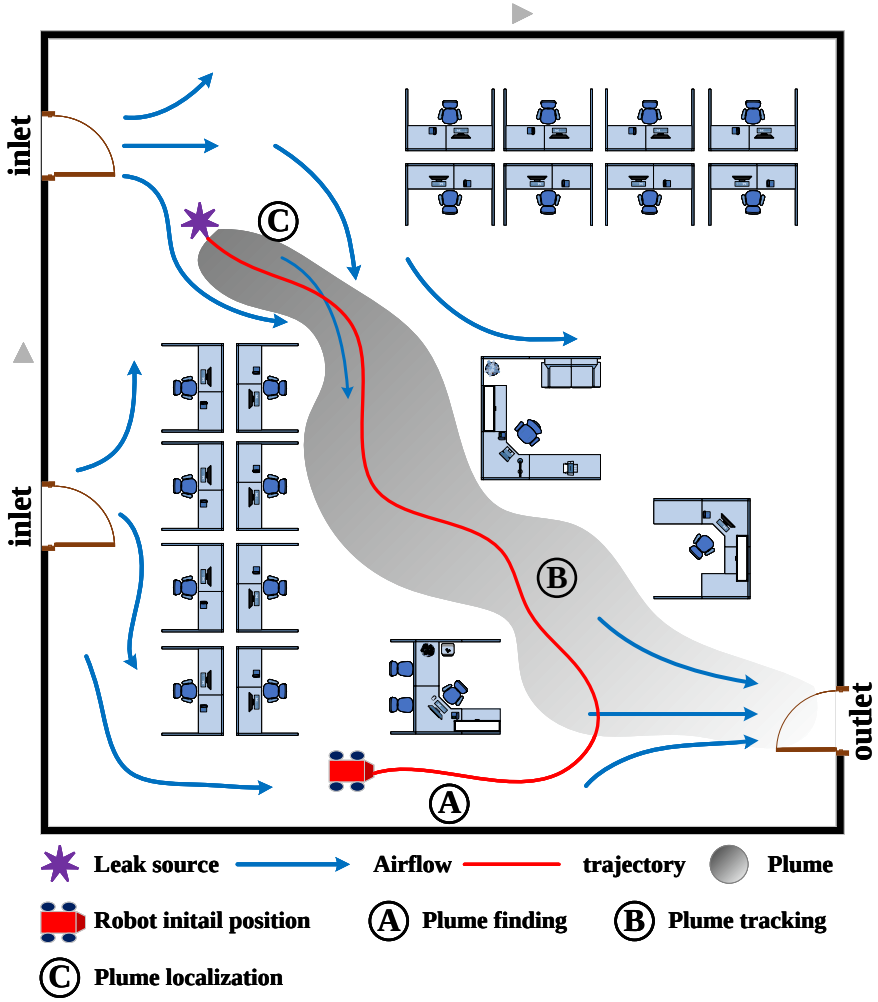



Figure 1. Illustration of the odor source localization task. The robot performs a random search to enter the plume (Phase Ⓐ), navigates toward high-concentration regions (Phase Ⓑ), and declares the leak source location (Phase Ⓒ).

tial decision-making problem characterized by a high-dimensional state space and sophisticated action requirements. By leveraging deep neural networks to extract features from sparse data, DRL is able to optimize the decision-making process and has therefore been applied to OSL tasks. Multiagent DRL (MADRL) [9] is a paradigm that extends DRL to the cooperation or competition of multiple agents within a shared environment. It is commonly adopted in autonomous driving, unmanned aerial vehicle (UAV) swarms, and Internet of Things (IoT) systems. When single-agent DRL is extended to the multi-agent setting, it suffers from the curse of dimensionality and the attendant computational complexity [10]: as the number of agents increases, the joint state space and the joint action space grow exponentially. Meanwhile, environment non-stationarity [11] subjects each agent to the continuously evolving policies of the other agents, making stable learning and convergence difficult to achieve. Existing MADRL-based OSL studies, however, do not take complex environments with obstacles into account. Moreover, in non-stationary environments, agents are confronted with sparse reward signals: prolonged exploration is required before valid signals can be obtained, which gives rise to the exploration-exploitation dilemma [12].

To effectively address the aforementioned challenges, this article proposes a diffusion model-enhanced MADRL algorithm. Specifically, we exploit the generative capability of diffusion models (DM) to enhance the decision-making capability of the policy network, thereby establishing an effective framework for OSL tasks. Our main contributions are summarized as follows:

(1) Dec-POMDP formulation with a composite reward design. To capture the dynamic behavior of a time-varying source characterized by periodic emission and attenuation, we formulate the OSL task as a decentralized partially observable Markov decision process (Dec-POMDP) with unknown transition probabilities. A multi-agent state space and a composite reward function are further designed to alleviate the challenge posed by sparse rewards.

(2) DM-enhanced MADRL algorithm. We propose a diffusion model-enhanced multiagent proximal policy optimization (DM-MAPPO) algorithm, incorporating the reverse denoising process of DM into the policy network. Specially, we exploit the inference and generative capabilities of DM to achieve high-quality decision-making.

(3) Extensive simulation validation. Extensive simulations are conducted to verify the effectiveness and superiority of the proposed algorithm. The results demonstrate that DM-MAPPO outperforms existing baseline algorithms.

The remainder of this article is organized as follows. Section II reviews related works. Section III presents the system model and problem formulation. Section IV details the proposed method and algorithm. Simulation results are presented in Section V, and the conclusion is provided in Section VI.

## 2. Related work

In this section, we summarize the contributions of related work and highlight the distinctions between prior studies and the proposed method.

### *A. Bio-Inspired Heuristic and Probabilistic Inference Methods*

Regarding bio-inspired algorithms, Lin et al. [13] drew inspiration from the collective hunting behavior of whales in nature. They improved the shark smell optimization (SSO) algorithm by introducing a collective information-sharing mechanism and a variance-triggered mechanism for escaping local extrema, thereby developing the I-SSO algorithm. This design prevents robots from becoming trapped in local concentration extrema during exploration. Xu et al. [14] proposed a model-free Zigzag-Spiral algorithm: the coarse search stage enables the target signal to be detected, whereas the fine search stage adopts a novel spiral search strategy to improve localization accuracy.

Inverse methods based on probabilistic inference have also attracted considerable attention. Vergassola et al. [15] proposed a probabilistic, information-driven strategy termed Infotaxis, which exploits information entropy to guide a robot toward an odor source. Recent studies have sought to integrate Entrotaxis with complementary techniques. Zhao et al. [16] combined Entrotaxis with a regression method and developed regression-enhanced Entrotaxis. A reward function derived from the maximum entropy sampling principle is used to guide next action. Guo et al. [17] proposed Clutaxis, an information-driven search method based on a global exploration-exploitation tradeoff principle. It achieves higher computational efficiency and yields more flexible search trajectories in turbulent environments.

Nevertheless, both methods yield satisfactory performance only under specific environmental conditions. When the deployment scenario changes, the parameters of robots must be redesigned to accommodate the new environment, which renders these methods inflexible. Moreover, because such methods lack the ability to perceive temporal characteristics, their performance degrades in environments with sparse and intermittent plumes. In addition, the effective resources available to the robots are typically limited, which prevents the above methods from maintaining stable performance in complex environments [18].

### *B. MADRL with Diffusion Model for the OSL Task*

MADRL enables multiple agents to pursue distinct optimization objectives and therefore exhibit significant advantages in high-dimensional and continuous plume environments. Alagha et al. [19] proposed two novel MADRL models that combine MAPPO, a convolutional encoder, and team reward shaping, thereby achieving efficient cooperative localization in wall-containing

environments. Policy learning is further guided by expert demonstration cloning, which reduces the training time while preserving localization performance. Alarcon Granadeno et al. [20] introduced an Action-Specific Double Deep Recurrent Q-Network (ADDRQN) to apply MADRL to three-dimensional multi-source plume tracing. The network conditions on action-observation pairs and integrates teammate information through a permutation-invariant set encoder, thereby accomplishing multi-agent cooperation. However, MADRL-based OSL methods rely on historical experience, require extensive environment interactions, and suffer from difficult hyperparameter tuning in sparse-reward settings. As the number of agents increases, the learning cost grows accordingly. This may prevent the DRL algorithms from making optimal decisions based on the current state.

In recent years, the advent of generative artificial intelligence (GAI) has drawn growing attention to its powerful generative capabilities. Among the resulting models, denoising diffusion probabilistic model (DDPM) [21] and denoising diffusion implicit model (DDIM) [22] exhibit remarkable generative performance and have been adopted across a wide range of domains. In audio and text generation, DM is employed to synthesize acoustic symbols and textual content. In computer vision, it can generate high-quality images and video from textual descriptions. More importantly, DM has demonstrated superior performance on decision-making problems, thereby offering a new perspective for addressing the OSL problem.

## 3. System model

In this section, we first present the gas dispersion model and then formulate the OSL task as a decentralized partially observable Markov decision process with unknown transition probabilities.

### *3.1. Plume dispersion model of OSL task*

This article employs the Fluent module in ANSYS to simulate the process of plume dispersion in a 2D indoor ventilation environment. The environment with obstacles is defined as a $10 \times 10$ m square domain, featuring two inlets (each 1m wide) on the left boundary and a single 1 m-wide outlet on the right boundary. The plume leak source, modeled as a $0.1 \times 0.1$ m square, is positioned at coordinates (3,7) in the domain and emits plume along the negative y-axis direction, as shown in Figure 2. Three square obstacles (each with a 1 m side length) and two rectangular obstacles measuring $3 \times 0.4$ m are introduced to create more complex airflow conditions.

Considering the computational complexity of plume dispersion simulations in CFD and the difficulty of model training, the mesh resolution is set to 0.2 m, yielding 2349 mesh cells. This design reduces the computational complexity while preserving the model accuracy.

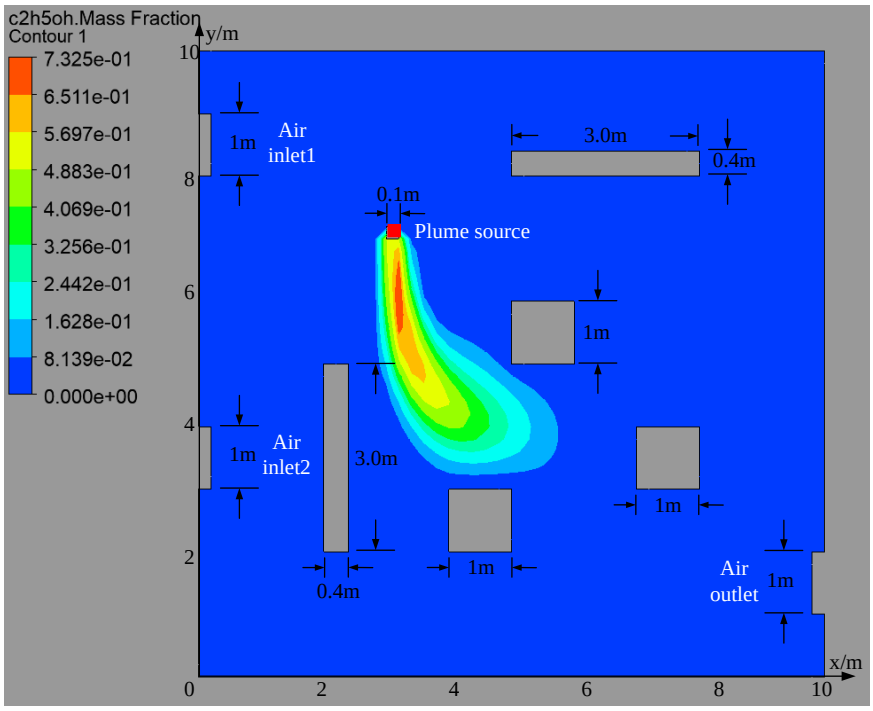



Figure 2. The indoor environment with obstacles.

In this work, the Reynolds-averaged Navier-Stokes (RANS) formulation with the standard $k-\epsilon$ turbulence model is employed to simulate the dispersion process. Since plume dispersion in an indoor environment with airflow exhibits an irregular spatial distribution, this computational model is well suited to the complex indoor environment. To better reproduce the plume release in realistic scenarios, we superimpose a periodic time-varying component on the decaying release rate, as expressed by:

$$v(t) = v_0(1 + k\sin(2\pi f t))e^{-t/40} \tag{1}$$

where $v(t)$ is the time-varying release rate of the plume source, $v_0$ represents the initial release rate, $k$ is the dimensional fluctuation ratio, and $f$ is the fluctuation frequency. The boundary conditions and solution models for CFD simulation are shown in Table 1.

Table 1. Boundary Condition and Solver Settings for CFD.

| Boundary type | Configuration | Category | Model/Methods |
|---|---|---|---|
| Source | Mass flow inlet | Vision model | RANs $k-\epsilon$ model [23] |
| Obstacle | No-slip wall | Species model | Species transport |
| Wall | Zero diffusive flux | Mixture material | Air and alcohol |
| Air inlet | Velocity inlet | Velocity-pressure coupling | SIMPLE method |
| Air outlet | Relative pressure outlet | Time type | Transient |

To improve the computational tractability of the CFD simulation, we model the search area as a square grid. Based on the prescribed grid size, the environment is divided into $M$ grid cells, as shown in Figure 3. The vector $C = [c_1, c_2, c_3, \ldots, c_M]$ defines the position of the robot within the grid, where $M = n \times n$, and $n$ denotes the number of cells per row and column. In addition, $c_i = (x_i, y_i)$ represents the search location of the robot in the environment, where $i \in \{1, 2, \ldots, M\}$. Since the location of the plume leakage source is unknown to the robot, any one of the $M$ grid cells may potentially be the source position.

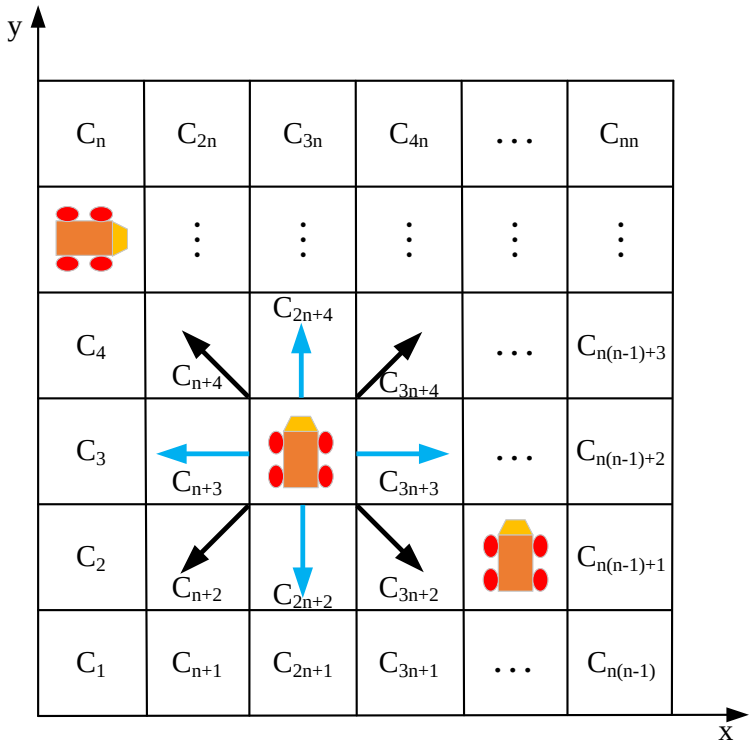



Figure 3. Search area and eight motion directions.

### *3.2. Modeling the OSL task as the Dec-POMDP*

In the OSL task, Robots' states are determined by the information measured by their onboard sensors. Robots' actions are independent of the history and depend only on the current state, so the next actions are chosen according to the currently extracted information. This process can be modeled as a decentralized partially observable Markov decision process (Dec-POMDP) with unknown transition probabilities [24]. Generally, a Dec-POMDP can be represented as the six-tuple $\langle S, A, P, R, O, \gamma \rangle$, where $S$ is the global state space of the environment, $A$ is the action space of the robots, $P$ denotes the state transition probability of the environment, $R$ is the reward space,

$O$ is the local observation space, and $\gamma \in [0, 1]$ is the reward discount factor. The Dec-POMDP with unknown transition probabilities is represented by the tuple $\left(s_t^n, a_t^n, R_t^n, s_{t+1}^n, n\right)$, where $n$ is the index of the robot.

In the OSL task, the robots as the agents of the Dec-POMDP implement sequential decision-making through interaction with the environment. The state of robot $n$ at any time $t$ is $s_t^n$, where $s_t^n \in S$. Then, the robot selects an action $a_t^n \in A$ according to the policy $\pi(s_t^n)$, thereby obtaining a reward $r_t^n$. The state $s_t^n$ transits to the next state $s_{t+1}^n$ based on the state transition probability function $P(s_{t+1}^n | s_t^n, a_t^n)$. After repeating the above process, the robot obtains the cumulative reward $\sum_{t=0}^{\infty} \gamma_t r_t^n$. $\gamma_t$ prompts the robot to select actions according to the optimal policy during decision-making, so that the expected cumulative reward $E\left[\sum_{t=0}^{\infty} \gamma_t r_t^n\right]$ is maximized. The basic elements of the Dec-POMDP for the OSL task are described as follows:

*A. State Space* In the OSL task, the plume concentration in an indoor environment varies with time, which requires the robots to perform periodic sampling to obtain environmental perception information. This article sets that the sampling period of the robot is $t$, and the observation value is denoted as $S_t = \{s_t^1, s_t^2, \cdots, s_t^n\}$. The sampling time is $T \in [0, 100]$. The robot must localize the leakage source in 100 s; otherwise, the experiment is regarded as a failure. The state space of a single robot is expressed as:

$$s_t^n(i) = \begin{bmatrix} C_{i-(m-1)}[x, y, q], C_{i+1}[x, y, q], C_{i+(m+1)}[x, y, q], \\ C_{i-m}[x, y, q], C_i[x, y, q], C_{i+m}[x, y, q], \\ C_{i-(m+1)}[x, y, q], C_{i-1}[x, y, q], C_{i+(m-1)}[x, y, q] \end{bmatrix} \tag{2}$$

where $C_i$ denotes that the source-seeking robot is located in the $i$th grid cell of the environment, and $[x, y, q]$ is the position information and the gas concentration $q$.

*B. Action Space*

The action set of a single robot is $A^n = \{a_t^1, a_t^2, \cdots, a_t^i\}$, where $a_t^i$ denotes 8 basic actions, as shown in Figure 3. The robot moves along the four directions, including up, down, left, and right with a fixed step size of $N$, while the four diagonal directions of up-left, down-left, up-right, and down-right have a fixed step length of $\sqrt{2}N$. The actions of all robots are concatenated to obtain the joint action space $A_t = \{A_t^1, A_t^1, \cdots, A_t^n\}$.

*C. Reward Function*

The reward $R$ describes the instantaneous reward obtained by the robot from the environment

when executing action $a$ in state $s$. $R$ is used to guide the robot to make correct decisions in the interaction with the environment. In the OSL task, robots mainly rely on concentration information to find the leakage source. Therefore, this article designs the reward based on concentration information, mainly considering the following points: (1) multi-agent cooperation to solve the high sparse reward problem in the OSL task; (2) efficiently exploring the plume concentration distribution area; (3) avoiding crossing boundaries, colliding with obstacles, and collisions among robots; (4) completing the OSL task with the minimum number of search steps.

For each robot, exploring the plume concentration is always the objective. Therefore, the reward based on plume concentration is divided into concentration exploration reward $R_1$, concentration exploitation reward $R_2$, declaration reward $R_3$, obstacle avoidance reward $R_4$, time step penalty $R_5$. $R_1$ is designed to rapidly find the plume. $R_2$ encourages robots to find the plume source based on concentration information. $R_3$ is assigned to penalize behaviors of boundary violations and obstacle collisions. $R_4$ penalizes robots that collide with each other. $R_5$ encourages the robot to find the leakage source with fewer search steps. The reward functions are formulated as follows:

$$
\begin{aligned}
R_1 &= \begin{cases} 0.5, & \mathbf{q}_{t+1} \geq q_t \\ 0, & \mathbf{q}_{t+1} < q_t \end{cases}, \quad R_2 = \begin{cases} 2\left(q^i_{\max} - q_t\right), & \mathbf{q}_{t+1} \geq q_t \\ q^i_{\max} - q_t, & \mathbf{q}_{t+1} < q_t \end{cases}, \\
R_3 &= \begin{cases} -300, & \text{beyond the boundaries} \\ -10, & d < d_{\text{tm}} \\ 300, & \mathbf{q}_t \geq q_{\max} \end{cases}
\end{aligned} \tag{3}
$$

$$
R_4 = -300, if c_i = c_j; R_5 = -0.1 \tag{4}
$$

where $q^i_{\max}$ is the maximum concentration among the eight directions; $q_t$ represents the concentration at the robot's own location at time $t$. $c_i$ and $c_j$ denote the positions of any two robots.

The mathematical form of the cumulative reward $R_{total}$ under $k$ search steps is:

$$
R_{total} = \sum_k \left(R_1^k + R_2^k + R_3^k + R_4^k + R_5^k\right) \tag{5}
$$

## 4. OSL task algorithm based on DM-MAPPO

In this section, the diffusion model is introduced, and a diffusion model-enhanced MAPPO algorithm is subsequently proposed.

### *4.1. DM-enhanced decision making*

*A. Motivation* Based on the characteristics of the OSL task and the advantages of diffusion models in decision-making, the reasons why this article uses DM to enhance the decision-making capability of robots are summarized as follows:

(1) Diffusion models can generate high-quality and diverse action samples. In current research, datasets provided by expert systems are used to improve the adaptive capability of robots in dynamic environments [25][26]. However, in the OSL task, plume dispersion is highly dynamic and time-varying, which makes empirical data difficult to reuse. DMs have become an ideal tool for solving this problem. DMs can exhibit strong generative capability under dynamic environmental conditions through online learning and offline inference to solve the OSL problem.

(2) Guided by wind information, the plume disperses. Upon collision with obstacles, stochastic dispersion is further induced, even forming local extreme-value regions of concentration in corners. This leads to the high dynamicity of the environmental perception information obtained by the robot in indoor ventilation environments. Therefore, the modeling and description of the OSL task is a typical complex, high-dimensional problem. Fortunately, DMs can model complex data distributions and have strong generalization capability in highly dynamic indoor environments and task modes. In addition, the OSL task involves concentration information, wind information, obstacle information, and unsensed regions. The information perceived in different regions differs from each other; they are both independent and complementary. Diffusion models can identify the features of sensed and unsensed regions and model the complex interactions among different factors, thereby achieving more effective decision-making in the OSL task.

(3) DMs can enhance the decision-making capability of DRL in complex and high-dimensional environments. DRL uses multilayer perceptron as function approximators, which faces the problem of difficulty in capturing dependency relationships in complex data. Especially in complex and dynamic environments, DRL suffers from inefficient learning and poor generalization. DMs can use the data learning capability to complement and enhance DRL, efficiently improving the performance of the OSL task.

*B. Diffusion model-based optimal policy*

In the OSL task, the decision is represented as a set of discrete action probabilities. The environmental perception information and the mutual cooperation among robots are regarded as the dynamic environment. The diffusion model used in this article is DDPM. Specially, the forward process of DDPM continuously adds Gaussian noise to the optimal decision for the current environment, and the reverse process removes the noise according to the current environment to obtain

the optimal decision.

***Forward Process:*** Assume $x_0$ denotes the raw data of the decision based on the current environment. Gaussian noise is continuously added for $K$ steps, yielding a series of noisy samples $\{x_k\}_{k=0}^{K}$. Specifically, for the discrete vector $x_{k-1}$ at the $k-1$ step, Gaussian noise is added to obtain the noisy data $x_k$. The Gaussian sampling $q(x_k|x_{k-1})$ is expressed as:

$$q(x_k \mid x_{k-1}) = \mathcal{N}\left(x_k; \mu_k = \sqrt{1-\beta_k}x_{k-1}, \Sigma_k = \beta_k \mathbf{I}\right) \tag{6}$$

where $\mathbf{I}$ is the identity matrix, $q(x_k|x_{k-1})$ follows a normal distribution, and $\beta_k$ is the variance of the forward process, calculated as follows:

$$\beta_k = 1 - e^{-\frac{\beta_{min}}{k} - \frac{2k-1}{2K^2}(\beta_{max}-\beta_{min})} \tag{7}$$

where $\beta_{\max}$ and $\beta_{\min}$ denote the maximum variance and the minimum variance, respectively, and $K$ denotes the number of diffusion steps.

Since $x_k$ follows the Markov process, the posterior probability from $x_0$ to $x_k$ is expressed as:

$$q(x_{1:K} \mid r_0) = \prod_{k=1}^{K} q(x_k \mid x_{k-1}) \tag{8}$$

Therefore, the mathematical relationship between the original data $x_0$ and the arbitrary data $x_k$ can be expressed as:

$$x_k = \sqrt{\bar{\alpha}_k}x_0 + \sqrt{1-\bar{\alpha}_k}\epsilon_k \tag{9}$$

where $\alpha_k = 1-\beta_k$, $\bar{\alpha}_k = \prod_{h=1}^{k} \alpha_k$, $\epsilon_k \sim N(0, \mathrm{I})$. It should be noted that, in the OSL task, the forward process of the diffusion model needs to add noise on the basis of obtaining the optimal decision, whereas the optimal decision is not available in the actual process. Therefore, this article does not consider it into the proposed method.

***Reverse Process:*** The purpose of this process is to reconstruct the initial data $x_0$ from the noisy data $x_k$. First, a parameterized model $p_\theta(x_{k-1}|x_k)$ is used to model the Gaussian distribution; then, the probability from $x_k$ to $x_0$ is as follows:

$$p_\theta(x_{0:K}) = p_\theta(x_K) \prod_{k=1}^{K} p_\theta(x_{k-1} \mid x_k) \tag{10}$$

$$p_\theta(x_{k-1} \mid x_k) = \mathcal{N}(x_{k-1}; \mu_\theta(x_k, k, s), \bar{\beta}_k \mathbf{I}) \tag{11}$$

where $p(x_K) = \mathcal{N}(0, \mathbf{I})$, the mean $\mu_\theta$ is learned by a deep neural network, the observation state $s$ is used as the denoising condition, and the deterministic variance magnitude $\bar{\beta}_k$ is expressed as:

$$\bar{\beta}_k = \frac{1 - \bar{\alpha}_{k-1}}{1 - \bar{\alpha}_k} \beta_k \tag{12}$$

According to Equation (9) and [27], the mean $\mu_\theta$ can be reconstructed as:

$$\mu_\theta(x_k, k, s) = \frac{1}{\sqrt{\alpha_k}} \left( x_k - \frac{\beta_k}{\sqrt{1 - \bar{\alpha}_k}} \epsilon_\theta(x_k, k, s) \right) \tag{13}$$

where $\epsilon \sim N(0,1)$, $k = 1, 2, \cdots, K$. $\epsilon_\theta(x_k, k, s)$ is trained through a neural network.

Then, sampling via the reverse denoising process and the reparameterization trick yields:

$$x_{k-1} \mid x_k = \frac{x_k}{\sqrt{\alpha_k}} - \frac{\beta_k}{\sqrt{\alpha_k(1 - \bar{\alpha}_k)}} \epsilon_\theta(x_k, k, s) + \sqrt{\beta_k} \epsilon \tag{14}$$

where $k = 1, 2, \cdots, K$ and $\epsilon \sim \mathrm{N}(0, I)$.

The loss function for training the denoising network model is expressed as follows:

$$\mathcal{L}_k = \mathbb{E}_{x_0, k, \epsilon} \left[ \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta \left( \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, t \right) \right\|^2 \right] \tag{15}$$

Finally, we introduce $x_0$ into the Softmax function, and obtain the probability distribution over the action space as:

$$\pi_\theta(s) = \left\{ \frac{e^{x_0^a}}{\sum_{k=1}^{A} e^{x_0^k}} \quad \forall a \in A \right\} \tag{16}$$

### *4.2. DM-MAPPO algorithm*

The structure of the DM-MAPPO algorithm proposed in this article is shown in Figure 4, and its pseudocode is given in Algorithm 1. DM-MAPPO adopts the training method of centralized training and decentralized execution (CTDE). Each robot makes decisions based on its own observation and reward, while the evaluation of the decision adopts the global state. In the OSL task, all robots search for the leakage source through cooperation. Therefore, robots need to share information to achieve a common goal by learning from both their own observations and the information provided by other robots.

MAPPO adopts the Actor-Critic architecture. In the network structure, each robot has an independent Actor network, and all robots share one Critic network. Then, the global Critic network optimizes the parameter $\pi_{\theta_t^n}$ through the policy function $\theta_t^n$. The minimized loss function is ex-

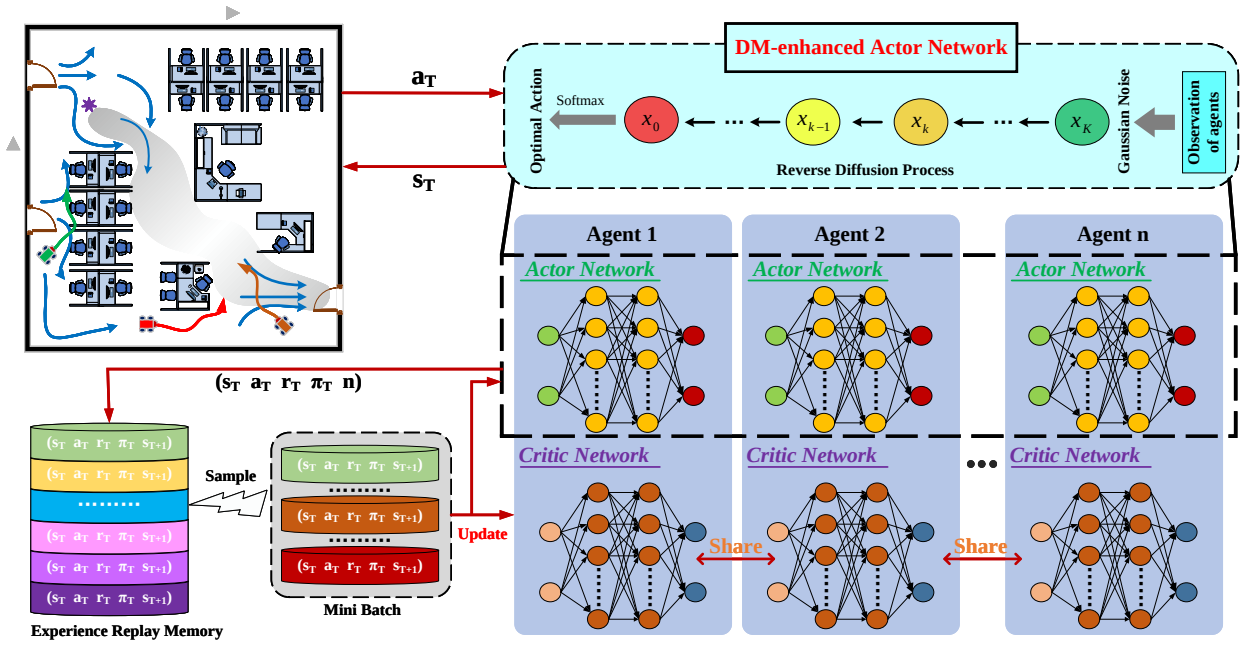



Figure 4. Framework of the proposed DM-MAPPO algorithm.

pressed as follows:

$$\theta_t^n = \arg\min \mathbb{E}\left[V_{\theta_t^n}^{\pi_t}(s_t) - \hat{A}_t^n(s_t, a_t) - V_{\theta_{old}^n}^{\pi_t}(s_t)\right]^2 \tag{17}$$

where $\hat{A}_t$ is evaluated by generalized advantage estimation (GAE) [28], expressed as follows:

$$\hat{A}_t^n = \sum_{l=0}^{h} (\gamma\lambda)^l \left(r_{t+l}^n(s_{t+l}, a_{t+l}) + \gamma V_{\phi^n}(s_{t+l+1}) - V_{\phi^n}(s_{t+l})\right) \tag{18}$$

To improve computational efficiency, MAPPO heuristically uses policy clipping and value clipping to restrict the value loss within a trust region. The problem is transformed into an unconstrained optimization problem, rather than constraining the KL divergence. It is expressed as follows:

$$L^{\text{CLIP}}(\pi_{\theta_t^n}) = \widehat{\mathbb{E}}_t\left[\min\left(r_t(\theta)\hat{A}_t^n(s_t, a_t), \text{clip}(r_t(\theta), 1-\xi, 1+\xi)\hat{A}_t^n(s_t, a_t)\right)\right] \tag{19}$$

where $r_t(\theta)$ is the importance sampling factor, expressed as follows:

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_t^n}(a_t \mid s_t)} \tag{20}$$

where *clip* is the clipping function; the smaller its value, the more cautious the policy update. $\xi$ is a hyperparameter.

**Algorithm 1** DM-MAPPO

Initialize the weights of the critic network $\theta$ and the weights of the actor network $\phi$
Initialize agents' DM policy network $\{\pi_\theta\}$ and value network $\{\phi\}$
Initialize experience replay memory $\mathcal{G}$ and mini batch $\mathcal{D}$
**for** $episode = 1, 2, \ldots, N$ **do**
  **for** $t = 1, 2, \ldots, T$ **do**
    **for** $k = 1, 2, \ldots, K$ **do**
      Observe state $\mathbf{s}_t$ as a condition of denoising network and initialize a random normal distribution $\mathbf{x}_t \sim \mathcal{N}(0, \mathbf{I})$
      utilize a denosing network $\epsilon_\theta(x_k, k, s)$
      Obtain the reverse transition distribution $p_\theta(x_{k-1} \mid x_k)$ via Eq. (11) based on the mean by Eq. (12) and variance by Eq. (13)
      Calculate the distribution $\mathbf{x}_{k-1}$ via Eq. (14)
      Calculate the distribution $\mathbf{x}_0$ according to Eq. (10) to obtain action $a$
      Execute action $a$ to obtain reward $r$ and next state $s'$
      Store the transition $\{s, a, r, s'\}$ into the reply memory $\mathcal{G}$
      Randomly sample a mini-batch-sized transitions $\mathbf{T}_d$ from $\mathcal{G}$
      Deploy generalized advantage estimation to evaluate the advantages of $\mathbf{T}_d$
      Update the policy network $\{\pi_\theta\}$ according to Eq. (16)
      Update the value network $\{\phi\}$ according to Eq. (17)
    **end for**
  **end for**
**end for**
Return the policy network $\{\pi_\theta\}$ with the parameter $\theta$

## 5. Simulations

This section aims to evaluate the performance of the DM-MAPPO algorithm in the OSL task. This article provides the simulation setup and validates MAPPO performance in a dynamic environment. Following that, comparison and comprehensive analysis with other baseline algorithms are conducted.

### *5.1. Experiment setup*

*(1) Experimental Environment:* The coordinate of the leakage source is (3, 7). The size of one grid cell is the fixed step length of the robot's straight-line motion, and the fixed step length when the robot moves diagonally is 0.28 m. The detection interval between each action step is 3 seconds. Since the execution speed of the action is very fast, the robot's response execution time is neglected. In the training phase, all hyperparameters and empirical values related to the model are shown in Table 2. The implementation framework of the model is PyTorch, the constructed experimental environment is based on Gym, and the experimental environment is Xeon Silver 4510 3.9GHz, NVIDIA GeForce 3090, Windows 10, Pytorch 1.12.0+cu113, and Python 3.9.

Table 2. Hyperparameters and training thresholds of DM-MAPPO.

| Parameter | Value |
|---|---|
| Learning rate | 0.005 |
| Hidden width | 64 |
| Entropy regularization coefficient | 0.01 |
| Batch size | 512 |
| K-epochs (Number of data reuse times) | 20 |
| Discount factor | 0.99 |
| Maximum step per episode | 150 |
| Number of denosing step | 4 |
| Maximum $C_2H_5OH$ mass fraction | 0.99 |
| Concentration threshold for plume finding | 0.05 |
| Concentration threshold for declaring source | 0.6 |

*(2) Model Description:* The Actor network of the DM-MAPPO algorithm consists of three fully connected layers, and the hidden layer consists of 64 neurons activated by the ReLU function. The initial noise is sampled from a standard Gaussian distribution, and after multi-step reverse denoising, the discrete action probability distribution is output through the softmax function. The denoising step uses sinusoidal positional encoding with a dimension of 64. The Critic network adopts a three-layer MLP structure, and the hidden layer consists of 64 neurons activated by the

Tanh function. The Adam optimizer is used to update the Actor and Critic networks. The experience buffer adopts the mini-batch update method, and the batch size is 16.

*(3) comparison algorithms:* To evaluate the performance of the proposed DM-MAPPO on the OSL task, we compare it with the following benchmarks:

*MAPPO* [29]: is an extension of PPO [30] to multi-agent scenarios, applicable to both discrete and continuous action spaces. MAPPO adopts a centralized training and distributed execution framework, and is commonly used in multi-agent cooperation. This article selects MAPPO as a baseline algorithm to compare.

*MADQN* [31]: is an extension of deep Q-network (DQN) [32] to multi-agent scenarios, mainly used for multi-agent decision-making problems with discrete action spaces. Each agent shares the Q-network parameters, takes local observations as input, outputs the Q-values of each discrete action, and is trained through experience replay and target networks. DQN is one of the most traditional and widely used DRL algorithms in discrete action spaces. It has shown superior performance in solving various optimization problems. Therefore, this article selects MADQN as a baseline algorithm.

*VDN* [33]: is a MARL method based on linear value decomposition, applicable to discrete action spaces. VDN does not require a complex mixing network; instead, it simply uses the sum of individual Q-values as the joint Q-value, thereby simplifying training.

*QMIX* [34]: is a MARL method based on nonlinear value decomposition, designed for fully cooperative multi-agent tasks and applicable to discrete action spaces. QMIX adopts the CTDE framework. A centralized critic network receives the global state and guides the updates of the actor networks. Through a global Q-value function, QMIX governs the individual agents with the aim of maintaining policy consistency.

### *5.2. Results and Analysis*

*(1) Influence of DM-MAPPO Parameters:* The number of denoising steps is an important parameter of DM, so we have studied its influence on the average reward and the average number of search steps. To better compare the influence of the parameter on algorithm performance, we select the average reward and average number of search steps of 500 episodes in the steady state as a reference, as shown in Figure5 . Obviously, insufficient denoising steps lead to a very small reward value of the DM-MAPPO algorithm. Although a relatively large reward is obtained when N=2, the number of search steps is very large. This is because the discretization error increases in the reverse denoising process, which increases step-skipping during data sampling, resulting in a decrease in the quality of the generated actions and making it difficult for the robot to obtain a correct policy.

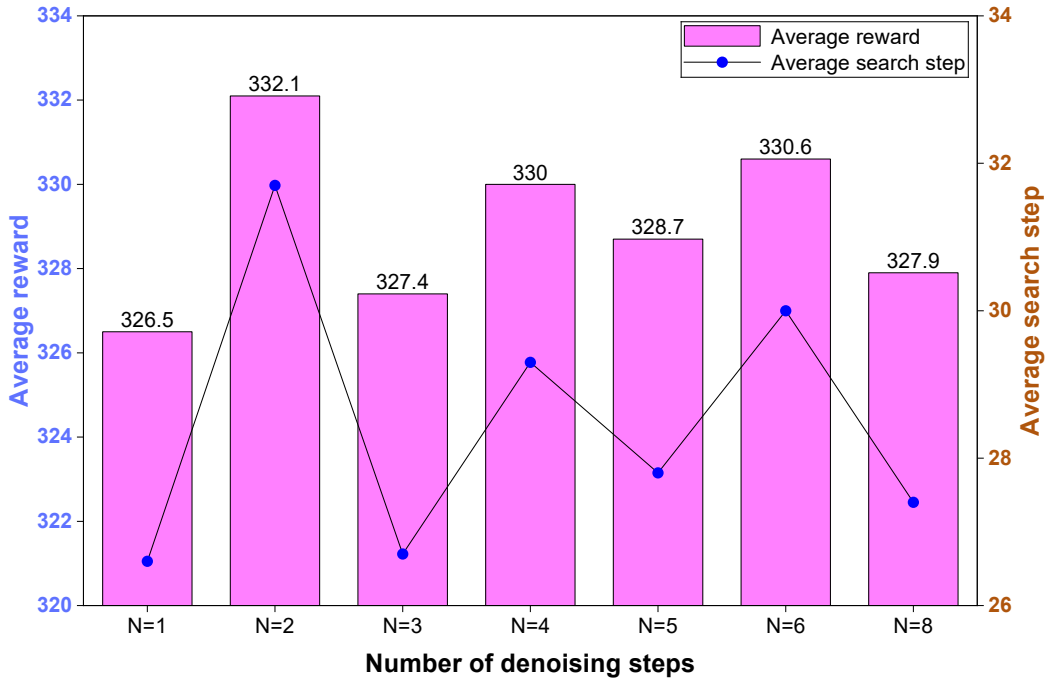



Figure 5. Average reward and search steps of DM-MAPPO algorithm on different denoising steps

However, when the number of denoising steps is excessively large, the reward of the DM-MAPPO algorithm exhibits a significant decline. This degradation arises because model prediction errors accumulate at each step, which restricts the diversity of the actions generated by the algorithm and consequently hinders the exploration capability of the robots. Moreover, an excessively large number of denoising steps necessitates covering more noise levels during training, thereby incurring higher training costs and increasing the difficulty of evaluation and optimization in the Critic network. Therefore, the number of denoising steps is an important indicator for balancing exploitation and exploration in robots. Based on the above analysis and experimental results, we adopt a denoising step number of 4 to achieve favorable algorithmic performance.

Learning rate is an important hyperparameter of DRL algorithms and plays an important role in the convergence and stability of algorithm training. Therefore, we study the performance of the DM-MAPPO algorithm under different learning rates, as shown in Figure 6. The results show that a lower learning rate (L=0.001) leads to reduced sample utilization and makes it difficult to achieve the global optimum under the sparse reward in the OSL task. A higher learning rate (L=0.007) leads to excessively large policy updates, unstable algorithm performance, and insufficient exploration. Therefore, we set the learning rate to L=0.005 to achieve fast convergence and training stability.

*(2) Impact of Different Wind Speeds and Source Release Rates:* We use the obtained average

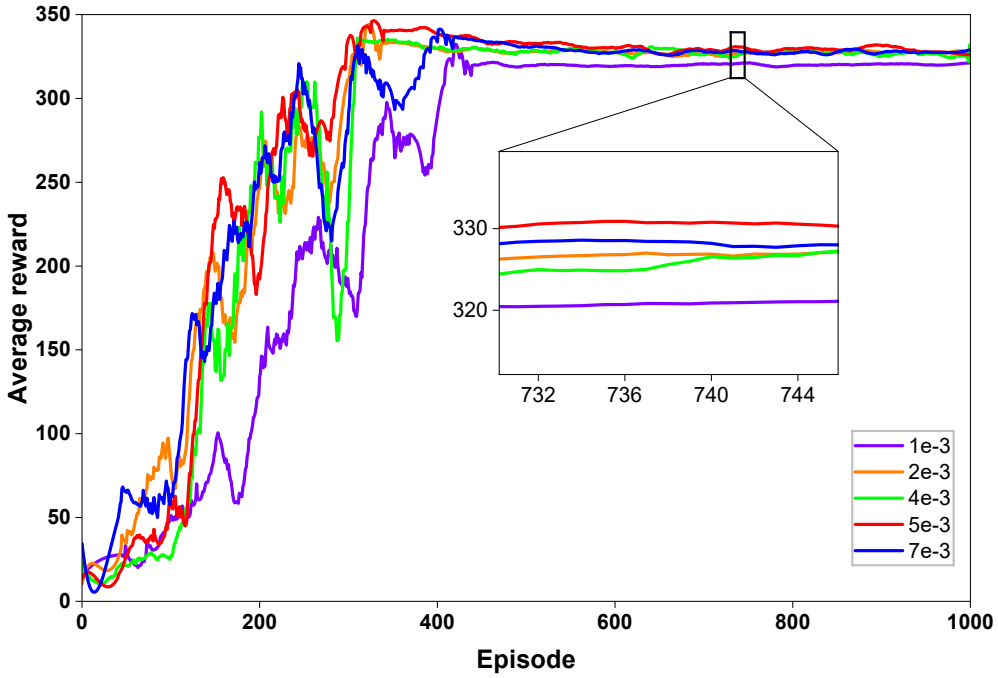
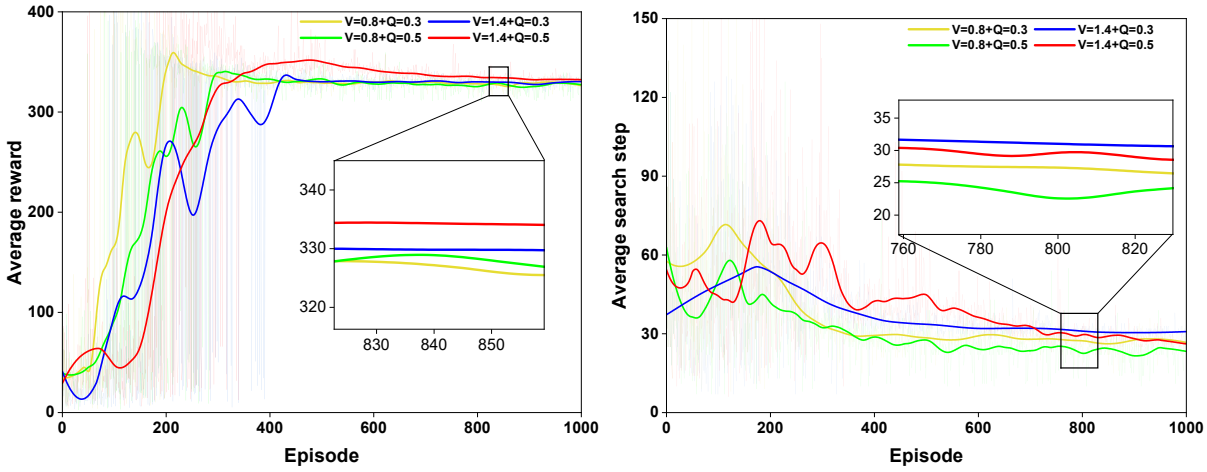



Figure 6. Convergence process of DM-MAPPO algorithm with different learning rates

reward and average number of search steps as the performance metrics of the plume source localization task, which conform to the rapidity and accuracy of completing the task. At the same time, the smaller the reward stable convergence episode and the average number of search steps, the higher the success rate, indicating better performance. This paper sets the positions of the 3 robots as (2.0, 1.0), (5.0, 1.0) and (8.0, 2.0), and the training results are shown in Figure 7. Guided by environmental perception, the robots can obtain more rewards. Table 3 shows that the increased wind speed increases the average value of the search steps and the standard deviation of the reward, while the increased concentration increases the average value of the reward. This is because the increase of wind speed and source release rate accelerates the plume diffusion, and local concentration extreme regions are prone to appear in the search area, causing the number of search steps to increase. Besides, the robots obtain information guidance from the distributed plume concentration, reducing the randomness of the search and obtaining more rewards. Therefore, the robots can complete the OSL task.

*(3) Comparison With Other MADRL Benchmarks:* We compare DM-MAPPO with other baseline algorithms, and the reward convergence curves and the number of search steps are shown in Figure 8. The best results of the statistical average reward and average number of search steps are given in Table 4. The results show that QMIX converges the slowest and obtains the least reward. VDN requires the most search steps to localize the leakage source. Both QMIX and VDN are




(a) Average reward

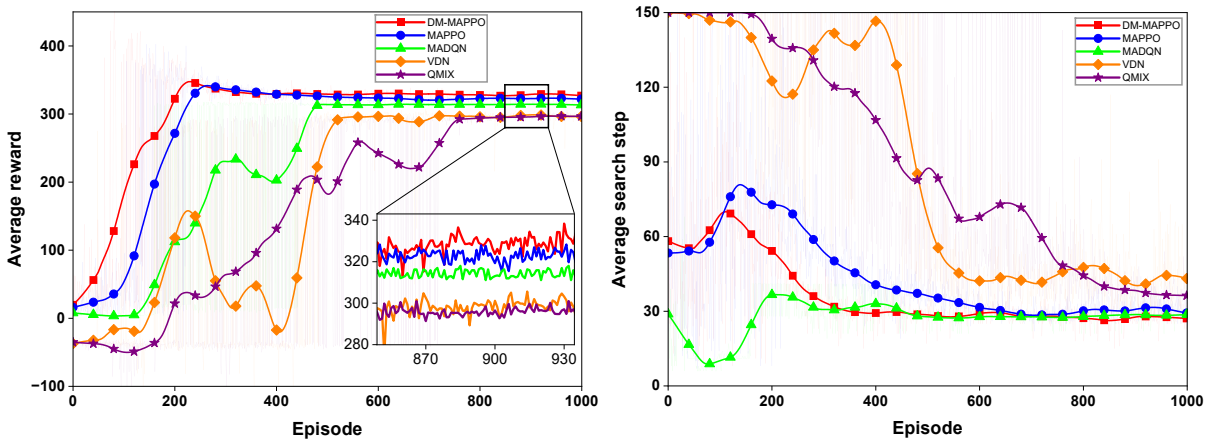
(b) Average search steps

Figure 7. Convergence process of DM-MAPPO under different wind speeds V and source release rate Q. The average reward is defined as the cumulative reward averaged over a series of episodes, while the average search step denotes the average number of steps taken by the robot across multiple experimental episodes.

Table 3. Search performance of proposed DM-MAPPO under different wind speeds V and source release rates Q.

| Wind speed + Source release rate | Average reward (standard deviation) | Average steps (standard deviation) |
|---|---|---|
| 0.8 m/s + 0.3 $m^3$/s | 329.01 ± 3.99 | 28 ± 3.23 |
| 0.8 m/s + 0.5 $m^3$/s | 328.27 ± 6.45 | 25 ± 5.72 |
| 1.4 m/s + 0.3 $m^3$/s | 330.13 ± 3.91 | 32 ± 2.64 |
| 1.4 m/s + 0.5 $m^3$/s | 339.54 ± 9.61 | 34 ± 8.38 |

value-decomposition based policies, and each robot makes decisions relying on its own observation, which makes the robots prone to falling into local optima in a sparse reward environment. At the same time, the complex cooperative relationships among robots cannot be exploited, leading to reduced exploration efficiency. MADQN obtains a lower reward, because its exploration capability in the OSL task is weak, resulting in insufficient policy optimization.




(a) Average reward

(b) Average search steps

Figure 8. Convergence comparison of reward and search steps between DM-MAPPO and benchmarks.

Table 4. Performance comparison of DM-MAPPO and benchmarks.

| Policy | Average reward (standard deviation) | Average steps (standard deviation) |
|---|---|---|
| MAPPO | 322.69 ± 2.97 | 30.74 ± 3.44 |
| MADQN | 313.92 ± 1.84 | 28.39 ± 2.31 |
| VDN | 295.58 ± 15.56 | 44.12 ± 12.73 |
| QMIX | 266.75 ± 90.21 | 38.34 ± 5.79 |
| DM-MAPPO | 328.83 ± 3.68 | 27.21 ± 3.03 |

Compared with other baseline algorithms, the proposed DM-MAPPO algorithm accomplishes the OSL task with higher rewards and fewer search steps. Compared with MAPPO, DM-MAPPO achieves a particularly pronounced performance improvement. This is attributed to the ability of the diffusion model to generate high-quality actions, which enables the robots to explore effectively in reward-sparse OSL tasks and thereby obtain higher rewards. In addition, benefiting from the high stability of MAPPO, DM-MAPPO achieves fewer search steps. Consequently, the proposed DM-MAPPO algorithm can effectively address the challenges in OSL tasks, exhibiting performance advantages such as fast convergence, strong stability, and a reduced number of search steps.

## 6. Conclusion

In this article, we studied the OSL task of multiple robots in complex scenes, using diffusion models to achieve optimal decision-making. We consider that the leakage source has dual time-varying characteristics of periodicity and decay, and describes it as a Dec-POMDP with unknown transition probabilities. To solve the reward sparsity problem of multiple robots in environments with obstacles, we design a multi-agent-based state space and a composite reward function to accelerate the learning process. Aiming at the problem of low-quality action selection caused by environmental changes, we use the generation and inference capabilities of diffusion models and applies the reverse denoising process to the policy network of the MAPPO algorithm to generate high-quality actions. Finally, a large number of experiments verify the effectiveness and superiority of the DM-MAPPO algorithm in dynamic environments. In future research, this method will be extended to larger indoor environments, and by integrating wind information and temperature into the reward function. We will also verify the effectiveness of DM-MAPPO in intermittent plume distribution environments.

### CRediT authorship contribution statement

**Qing Li:** Methodology, Formal analysis, Investigation, Writing - original draft, Project administration, Writing - review & editing. **Jie Yuan:** Data curation, Writing - review & editing, Funding acquisition, Software. **Ningning Zhang:** Investigation, Software. **Erkenbieke Jia:** Data curation, Funding acquisition.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.


### Acknowledgements

This research was funded in part by the National Natural Science Foundation of China under Grant 62263031, the Fundamental Scientific Research Funds for Universities in Xinjiang Uygur Autonomous Region under Grant 202532120001, and the Tianchi Project in Xinjiang Uygur Autonomous Region for Erkenbieke Jia.


### Data availability

Data will be made available on request.

## References


[1] C. Yu, Y. Dai, W. Wu, Z. Zou, C. Sun, H. Wang, Towards the optimization of pulsating ventilation system to improve leaked nitrogen dilution in a factory workshop by SIPC simulation, Energy and Buildings 357 (2026) 117202. doi:10.1016/j.enbuild.2026.117202.

[2] L. J. Molofsky, G. Etiope, D. C. Segal, M. A. Engle, Methane-rich gas emissions from natural geologic seeps can be chemically distinguished from anthropogenic leaks, Communications Earth & Environment 6 (2025) 11. doi:10.1038/s43247-024-01990-8.

[3] D. Hu, V. J. L. Gan, Semantic navigation for automated robotic inspection and indoor environment quality monitoring, Automation in Construction 170 (2025) 105949. doi:10.1016/j.autcon.2024.105949.

[4] S. Kim, J. Seo, H. Jang, C. Kim, M. Kim, J. Pyo, H. Oh, Gas Source Localization in Unknown Indoor Environments Using Dual-Mode Information-Theoretic Search, IEEE Robotics and Automation Letters 10 (2025) 588–595. doi:10.1109/LRA.2024.3511375.

[5] S. Ma, J. Yuan, Z. Guo, Q. Wu, Autonomous plume Near-Source search assisted by intermittent visible plume information using finite state Machine and YOLOv3-tiny, Expert Systems with Applications 228 (2023) 120350. doi:10.1016/j.eswa.2023.120350.

[6] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, D. Hassabis, Human-level control through deep reinforcement learning, Nature 518 (2015) 529–533. doi:10.1038/nature14236.

[7] K. H. Quah, C. Quek, Maximum reward reinforcement learning: A non-cumulative reward criterion, Expert Systems with Applications 31 (2006) 351–359. doi:10.1016/j.eswa.2005.09.054.

[8] M. Wang, B. Xin, M. Jing, Y. Qu, An Exploration-Enhanced Search Algorithm for Robot Indoor Source Searching, IEEE Transactions on Robotics 40 (2024) 4160–4178. doi:10.1109/TRO.2024.3454572.

[9] H. Kong, Q. Xing, Q. Wang, R. Niu, H. Chen, Y. Wang, S. Wang, Z. Duan, Y. Chang, ADAC: Actor-double-attention-critic for multi-agent cooperation in mixed cooperative-competitive

environments, IEEE Transactions on Intelligent Transportation Systems 26 (2025) 9579–9592. doi:10.1109/TITS.2025.3562302.

[10] Y. Wang, Y. Wang, F. Tian, J. Ma, Q. Jin, Intelligent games meeting with multi-agent deep reinforcement learning: A comprehensive review, Artificial Intelligence Review 58 (2025) 165. doi:10.1007/s10462-025-11166-1.

[11] Y. Wang, J. Lei, F. Shang, Y. Li, A comprehensive survey of multi-agent deep reinforcement learning for wireless spectrum management, Neurocomputing 653 (2025) 131236. doi:10.1016/j.neucom.2025.131236.

[12] K. Singh, S. Chatterjee, M. Mariani, Applications of generative AI and future organizational performance: The mediating role of explorative and exploitative innovation and the moderating role of ethical dilemmas and environmental dynamism, Technovation 133 (2024) 103021. doi:10.1016/j.technovation.2024.103021.

[13] Q. Lin, S. Wu, S. Wu, H. Wang, J. Zhang, Development and simulation of two novel indoor odor source localization methods using a modified shark smell optimization algorithm, Measurement 240 (2025) 115562. doi:10.1016/j.measurement.2024.115562.

[14] T. Xu, X. Sun, C. Wu, D. Ding, Model-free Zigzag-Spiral search for gas source localization in uncertain environments, Measurement 281 (2026) 121852. doi:10.1016/j.measurement.2026.121852.

[15] M. Vergassola, E. Villermaux, B. I. Shraiman, 'Infotaxis' as a strategy for searching without gradients, Nature 445 (2007) 406–409. doi:10.1038/nature05464.

[16] Y. Zhao, B. Chen, Z. Zhu, F. Chen, Y. Wang, Y. Ji, Searching the diffusive source in an unknown obstructed environment by cognitive strategies with forbidden areas, Building and Environment 186 (2020) 107349. doi:10.1016/j.buildenv.2020.107349.

[17] R. Guo, Y. Zhao, Y. Ji, M. Yan, Z. Zhu, Clutaxis: An information-driven source search method balancing exploration and exploitation in turbulent environments, Journal of Safety Science and Resilience 6 (2025) 125–137. doi:10.1016/j.jnlssr.2024.09.001.

[18] H. Magalhães, L. Marques, Motion patterns of cognitive odour source search behaviours, Information Sciences 740 (2026) 123233. doi:10.1016/j.ins.2026.123233.

[19] A. Alagha, R. Mizouni, J. Bentahar, H. Otrok, S. Singh, Multiagent Deep Reinforcement Learning With Demonstration Cloning for Target Localization, IEEE Internet of Things Journal 10 (2023) 13556–13570. doi:10.1109/JIOT.2023.3262663.

[20] P. A. Alarcon Granadeno, T. Chambers, J. Cleland-Huang, Multi-source plume tracing via multi-agent reinforcement learning under common UAV-faults, Machine Learning with Applications 22 (2025) 100737. doi:10.1016/j.mlwa.2025.100737.

[21] J. Ho, A. Jain, P. Abbeel, Denoising Diffusion Probabilistic Models, in: Advances in Neural Information Processing Systems, volume 33, Curran Associates, Inc., 2020, pp. 6840–6851.

[22] J. Song, C. Meng, S. Ermon, Denoising diffusion implicit models, 2022. doi:10.48550/arXiv.2010.02502. arXiv:2010.02502.

[23] F. Menter, Two-equation eddy-viscosity turbulence models for engineering applications, AIAA Journal 32 (1994) 1598–1605. doi:10.2514/3.12149.

[24] H. Li, J. Yuan, H. Yuan, An Active Olfaction Approach Using Deep Reinforcement Learning for Indoor Attenuation Odor Source Localization, IEEE Sensors Journal 24 (2024) 14561–14572. doi:10.1109/JSEN.2024.3373610.

[25] Z. Wang, X. Jin, T. Zhang, J. Li, D. Yu, K. H. Cheong, C. L. P. Chen, Expert System-Based Multiagent Deep Deterministic Policy Gradient for Swarm Robot Decision Making, IEEE Transactions on Cybernetics 54 (2024) 1614–1624. doi:10.1109/TCYB.2022.3228578.

[26] L. Wang, S. Pang, J. Li, Olfactory-Based Navigation via Model-Based Reinforcement Learning and Fuzzy Inference Methods, IEEE Transactions on Fuzzy Systems 29 (2021) 3014–3027. doi:10.1109/TFUZZ.2020.3011741.

[27] S. Liang, M. Yin, W. Xie, Z. Sun, J. Li, J. Wang, H. Du, UAV-enabled secure data collection and energy transfer in IoT via diffusion-model-enhanced deep reinforcement learning, IEEE Internet of Things Journal 12 (2025) 13455–13468. doi:10.1109/JIOT.2024.3521496.

[28] J. Schulman, P. Moritz, S. Levine, M. Jordan, P. Abbeel, High-dimensional continuous control using generalized advantage estimation, https://arxiv.org/abs/1506.02438v6, 2015.

[29] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, Y. Wu, The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games, in: 36th Conference on Neural Information Processing Systems, NeurIPS 2022, November 28, 2022 - December 9, 2022, volume 35 of

*Advances in Neural Information Processing Systems*, Neural information processing systems foundation, New Orleans, LA, United states, 2022.

[30] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, 2017. doi:10.48550/arXiv.1707.06347. arXiv:1707.06347.

[31] A. Tampuu, T. Matiisen, D. Kodelja, I. Kuzovkin, K. Korjus, J. Aru, J. Aru, R. Vicente, Multiagent cooperation and competition with deep reinforcement learning, PLOS One 12 (2017) e0172395. doi:10.1371/journal.pone.0172395.

[32] V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, M. Riedmiller, Playing atari with deep reinforcement learning, 2013. doi:10.48550/arXiv.1312.5602. arXiv:1312.5602.

[33] P. Sunehag, G. Lever, A. Gruslys, W. M. Czarnecki, V. Zambaldi, M. Jaderberg, M. Lanctot, N. Sonnerat, J. Z. Leibo, K. Tuyls, T. Graepel, Value-decomposition networks for cooperative multi-agent learning based on team reward, in: 17th International Conference on Autonomous Agents and Multiagent Systems, AAMAS 2018, July 10, 2018 - July 15, 2018, volume 3 of *Proceedings of the International Joint Conference on Autonomous Agents and Multiagent Systems, AAMAS*, International Foundation for Autonomous Agents and Multiagent Systems (IFAAMAS), Stockholm, Sweden, 2018, pp. 2085–2087.

[34] T. Rashid, M. Samvelyan, C. Schroeder de Witt, G. Farquhar, J. Foerster, S. Whiteson, Monotonic value function factorisation for deep multi-agent reinforcement learning, Journal of Machine Learning Research (JMLR) 21 (2020) Paper No. 178, 51.